

Macroeconomics

Financial Markets

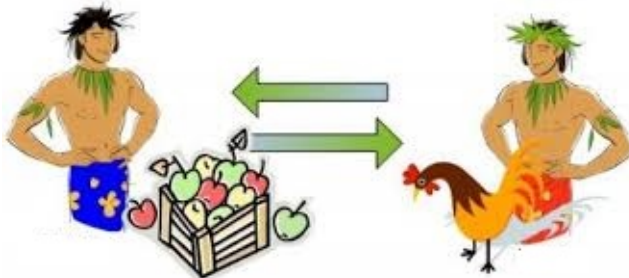
(Blanchard, Macroeconomics 8th edition Ch. 4)

Financial Markets

- ▶ Financial markets play an essential role in the economy.
- ▶ They involve a maze of institutions, from banks, to money market funds, mutual funds, investment funds, and hedge funds.
- ▶ Trading involves bonds, stocks, and other financial claims, such as swaps and options.
- ▶ The financial pages of newspapers quote interest rates on many government bonds, on many corporate bonds, on short-term bonds, on long-term bonds, etc.
- ▶ Financial markets determine the cost of funds for firms, households, government, and affect their spending decisions.
- ▶ We focus on role of the central bank in affecting interest rates.

Barter Economy

- ▶ Before the invention of money, services and goods are traded at negotiated rates.
- ▶ Requires double coincidence of wants.
- ▶ Too inefficient!



The Demand for Money

- ▶ Suppose only choose between two assets: money and bonds.
 - ▶ Money is used for transactions, but it pays no interest.
 - ▶ Types of money: currency, coins, bills, checkable deposits.
 - ▶ Bonds pay a positive interest rate, i (the rate of interest), but cannot be used for transactions.
- ▶ The holding of money and bonds depends on:
 - ▶ Level of transactions (because of some transaction costs).
 - ▶ Interest rate on bonds.
- ▶ You can hold bonds indirectly through funds
- ▶ In the early 1980s, the interest rate on US money market funds reached 14% per year, so people earned more interest by moving their wealth from checking accounts to these funds.

The Demand for Money

- ▶ Demand for money (M^d) is equal to nominal income $\$Y$ (a measure of level of transactions in the economy) times a decreasing function of the interest rate i :

$$\begin{matrix} \ominus \\ M^d = \$Y L(i) \end{matrix}$$

- ▶ An increase in the interest rate decreases the demand for money, as people put more of their wealth into bonds.
- ▶ The demand for money:
 - ▶ Increases in proportion to nominal income, and
 - ▶ Depends negatively on the interest rate.
- ▶ The relation between the demand for money and interest rate for a given level of income $\$Y$ is represented by the M^d curve.

M^d Curve

Who Holds US Currency?

- ▶ The amount of currency in circulation in 2006 was \$750 billion.
- ▶ US households together held \$170 billion in currency: average of \$1,600 per household
- ▶ Some was held by those involved in the underground economy or in illegal activities. When dealing with drugs, dollar bills, not checks, are the way to settle accounts.
- ▶ US firms held another \$80 billion.
- ▶ Foreigners abroad held \$500 billion, or 66% of the total, for transactions.

Who Holds US Currency? (con.)

- ▶ In a number of countries that have suffered from high inflation in the past, people have learned that their domestic currency may quickly become worthless and they see dollars as a safe and convenient asset.
- ▶ e.g., Argentina and Russia. Estimates suggest that Argentina holds more than \$50 billion in dollar bills, Russia more than \$80 billion, i.e., together close to the holdings of US households.
- ▶ In other countries, people who have emigrated to US bring home US dollar bills; or tourists pay some transactions in dollars, and the dollar bills stay in the country, e.g., Mexico, Thailand.

El Salvador and Venezuela

- ▶ A few countries (e.g., Ecuador, El Salvador) have actually adopted the dollar as their own currency. So people in these countries use dollar bills for transactions. But these countries are small.
- ▶ https://www.youtube.com/watch?v=LX5LCvooa_g
- ▶ https://www.youtube.com/watch?v=J_yCtg0buBU

Bitcoins

- ▶ Bitcoin: virtual assets that can be used in transactions, without third party being involved
- ▶ Total market value now around \$1 trillion (?), vs \$67 billion in December 2018, cf. US currency roughly \$2 trillion
- ▶ If bitcoins replace dollars, supply of currency would no longer be determined by the central banks, but by the mechanical rule that determines the creation of new bitcoins over time
- ▶ Not likely to happen:
 - ▶ Price of bitcoins fluctuates wildly, while prices still quoted in dollars
 - ▶ Electricity cost of verification and creation of bitcoins expensive
 - ▶ Political reasons: countries do not want to give up monetary policy

Bitcoins and El Salvador

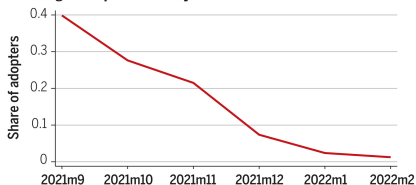
- ▶ <https://www.science.org/doi/10.1126/science.add2844>
- ▶ However, a currency's defining role is to serve as a medium of exchange, and cryptocurrencies have yet to be widely adopted as such.
- ▶ The government also launched an app called “Chivo Wallet,” which allows users to digitally trade both bitcoins and US dollars (USD, the official currency in El Salvador) without paying transaction fees
- ▶ the pandemic provided an additional incentive to adopt touchless payments

Bitcoins and El Salvador (con.)

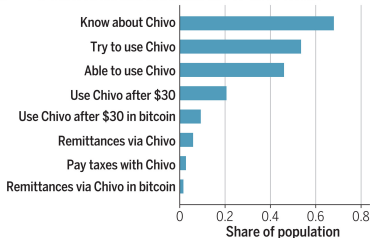
- ▶ bitcoin was not widely used as a medium of exchange and usage of Chivo Wallet was low.
- ▶ Most downloads took place just as the app was launched.
- ▶ Privacy and transparency concerns appear to be key barriers to adoption.
- ▶ technology involves a large initial adoption cost, has benefits that significantly increase as more people use it, and faces resistance from firms in terms of its adoption.
- ▶ firms accepting bitcoin were mostly large ... also more likely to accept cards.
- ▶ most firms reporting sales in bitcoin converted them into USD: ... only 12% of firms stored their sales in bitcoin within Chivo Wallet.

Bitcoins and El Salvador (con.)

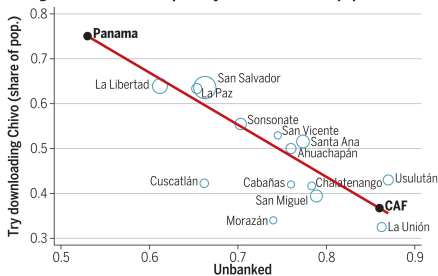
A Timing of adoption: Monthly downloads as a share of total downloads



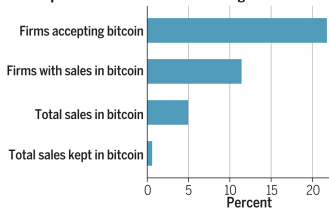
C Awareness and individual use of Chivo Wallet



B Regional variation in adoption by share of unbanked population



D Acceptance and use of bitcoin among firms



How Money Supply is Determined

- ▶ Central bank substantially controls quantity of money (money supply) that circulates in an economy.
- ▶ In US, the central banking system is the Federal Reserve System.
- ▶ Federal Reserve System directly regulates amount of currency in circulation (checking deposits, debit card accounts, other monetary assets).
- ▶ <https://www.theatlantic.com/business/archive/2011/04/the-destruction-of-money-who-does-it-why-when-and-how/236990/>

How Money Supply is Determined (con.)

- ▶ In Europe, it is the European Central Bank (ECB): <https://www.ecb.europa.eu/mopo/intro/role/html/index.en.html>
- ▶ “The central bank is the sole issuer of banknotes and bank reserves. That means it is the monopoly supplier of the monetary base. By virtue of this monopoly, it can set the conditions at which banks borrow from the central bank. Therefore it can also influence the conditions at which banks trade with each other in the money market.
- ▶ In the short run, a change in money market interest rates induced by the central bank sets in motion a number of mechanisms and actions by economic agents. Ultimately the change will influence developments in economic variables such as output or prices. This process – also known as the monetary policy transmission mechanism – is highly complex.”

How Money Supply is Determined (con.)

- ▶ Because DKK is pegged to EUR, Denmark cannot set money supply!
- ▶ Another example: Hong Kong, HKD is pegged to USD
- ▶ HK Monetary Authority: “Under a fixed exchange rate regime, while there is no independent monetary policy, money supply data can still be useful as an indicator of changes in economic activity such as output growth and inflation”.
- ▶ <https://www.youtube.com/watch?v=FdwPD2JbdJo>

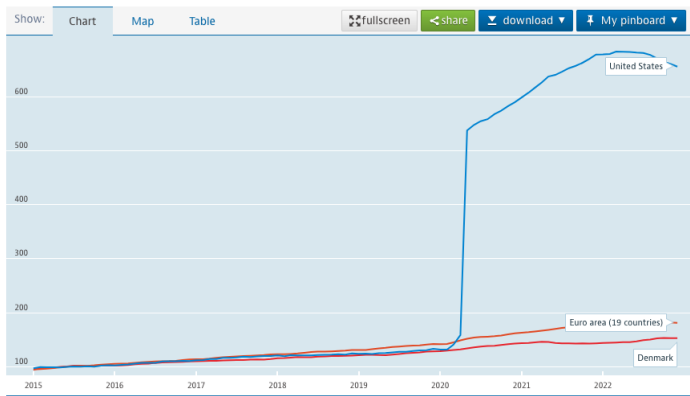
Definitions of Money Supply

- ▶ **Narrow money (M1)** includes currency i.e. banknotes and coins, plus overnight deposits.
- ▶ **Broad money (M3)** includes currency, deposits with an agreed maturity of up to two years, deposits redeemable at notice of up to three months and repurchase agreements, money market fund shares/units and debt securities up to two years.
- ▶ <https://data.oecd.org/money/broad-money-m3.htm#indicator-chart>

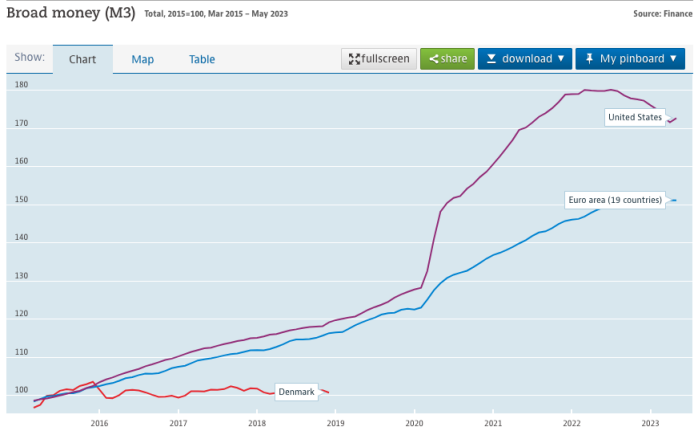
Narrow Money (M1)

Narrow money (M1) Total, 2015=100, Jan 2015 – Dec 2022

Source: Finance

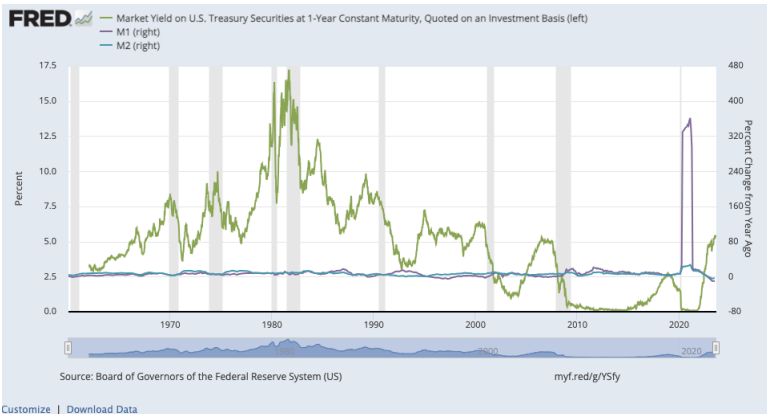


Broad Money (M3)



Increase in M1

- <https://fredblog.stlouisfed.org/2021/01/whats-behind-the-recent-surge-in-the-m1-money-supply/>



The Simple Story

- ▶ “Economists define money more narrowly as the component of wealth consisting of ‘transaction balances.’
- ▶ Because money is valued as a payment instrument, people are willing to hold a fraction of their wealth in money form for the sake of convenience, even though money earns relatively little interest and cash usually earns no interest at all.
- ▶ In late February and early March of 2020, the Fed cut its policy interest rate dramatically to help ease credit conditions during the COVID-19 crisis.
- ▶ The resulting acceleration in the supply of M1 can be understood largely as banks accommodating an increase in people’s demand for money.”

The More Complicated Side

- ▶ One factor responsible for this behavior may be related to a change earlier this year to Regulation D: The Federal Reserve requires banks to hold reserves against checkable deposits.
- ▶ But the regulation does not require banks to hold reserves against savings and money market accounts, which restrict depositors to no more than six transfers or withdrawals per month.
- ▶ On April 24, 2020, the Federal Reserve Board announced that Regulation D would no longer impose limits on the number of transactions or withdrawals permitted on savings deposit accounts.
- ▶ In any case, it seems that the modification of Regulation D in late April has effectively rendered savings accounts almost indistinguishable from checking accounts from the perspective of depositors and banks."

Danish Monetary Policy

- ▶ <https://www.nationalbanken.dk/media/ugtfuw hb/mon-pol-uk-09-web.pdf>
- ▶ Denmark has conducted a fixed-exchange-rate policy since the early 1980s, while several other small countries have opted for regimes based on explicit inflation targets since the early 1990s.
- ▶ Irrespective of the strategy chosen, the short-term interest rate is the primary instrument used in monetary policy.
- ▶ Due to the fixed-exchange-rate policy, the Danish monetary policy is aimed at keeping the krone stable against the euro. The main objective of monetary policy in the euro area is to maintain an inflation rate of below, but close to, 2 per cent.

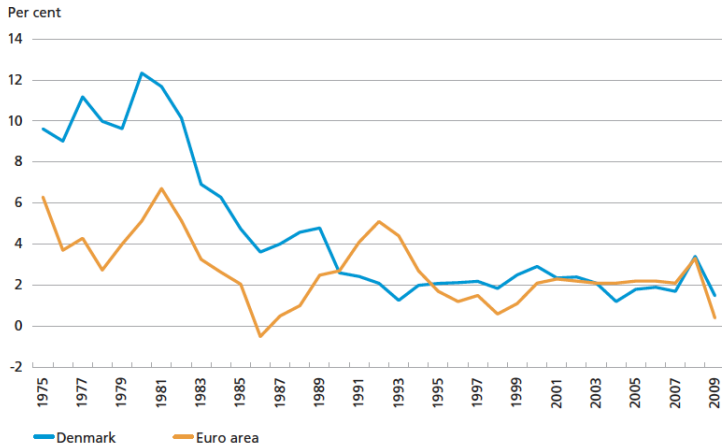
Danish Monetary Policy (con.)

- ▶ Over time, linking the Danish krone to the euro creates a basis for achieving the same level of inflation and inflation expectations in Denmark as in the euro area.
- ▶ If inflation is higher in Denmark than in the euro area, Denmark's competitiveness will, all other things being equal, deteriorate, leading to a loss of jobs in export sectors and in business sectors competing with imports.
- ▶ This dampens wage and price increases in Denmark, which eventually converge towards those of the euro area."

Convergence of Inflation

CONSUMER PRICE INFLATION 1975-2009

Chart 1.3



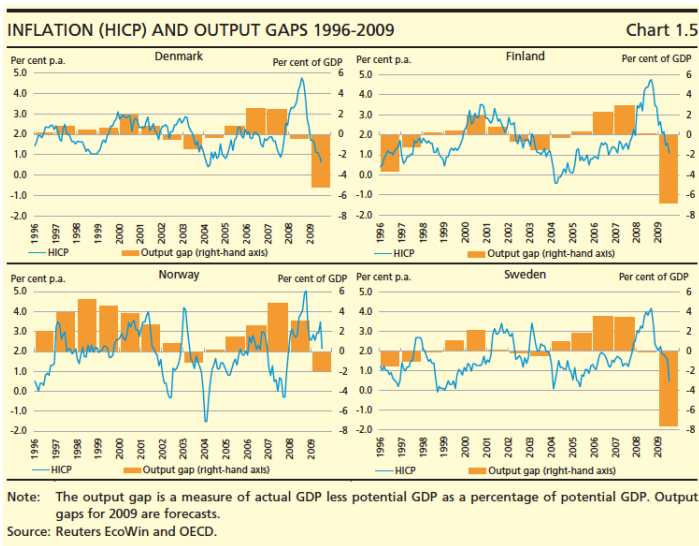
Note: Inflation in West Germany 1975-89 and Germany 1990-98. Preliminary annual averages for 2009.

Source: Statistics Denmark, ECB and OECD.

Comparison to Nordic Countries

- ▶ Since the mid-1990s, the Nordic countries have pursued different monetary-policy strategies.
- ▶ Denmark has conducted a fixed-exchange-rate policy against the D-mark and, from 1999, the euro.
- ▶ Finland switched to an inflation target in 1993. In 1996 it joined the ERM, and in 1999 it adopted the single European currency, the euro.
- ▶ Sweden has chosen to operate within the framework of an inflation target regimesince 1995.
- ▶ Norway operated with a trade-weighted exchange-rate index until 2001, when italso switched to an inflation target.

Comparison to Nordic Countries (con.)



Comparison to Nordic Countries (con.)

CONTINUED		Box 1.2			
INFLATION AND OUTPUT GAPS 1996-2009		Table 1.1			
		Denmark	Finland	Norway	Sweden
<i>Standard deviation</i>					
HICP, per cent year-on-year		0.77	1.03	1.15	0.92
Output gap, per cent of BNP		1.90	2.48	2.19	2.72
<i>Average</i>					
HICP, per cent year-on-year		2.01	1.67	1.93	1.60
Output gap, absolute level, per cent of BNP		0.83	1.17	3.88	1.03
Note: The output gap is a measure of actual GDP less potential GDP as a percentage of potential GDP. Output gaps for 2009 are forecasts.					
Source: Reuters EcoWin and OECD.					

Money Supply

- ▶ Suppose the central bank decides to supply an amount of money equal to M :

$$M^s = M$$

- ▶ Equilibrium in financial markets requires that $M^s = M^d = M$:

Money supply = Money demand

$$M = \$YL(i)$$

- ▶ The interest rate must be such that the supply of money (which is independent of the interest rate) is equal to the demand for money (which does depend on the interest rate).

Determining the Interest Rate

Comparative Statics

Monetary Policy, Open Market Operations

- ▶ Central banks typically change the supply of money by buying or selling bonds in the bond market, i.e., open market operations.
- ▶ *Expansionary* open market operation: the central bank expands the supply of money by buying bonds.
- ▶ *Contractionary* open market operation: the central bank contracts the supply of money by selling bonds.

The Balance Sheet of the Central Bank

Balance Sheet	
Assets	Liabilities
Bonds	Money (currency)

The Effects of an Expansionary Open Market Operation	
Assets	Liabilities
Change in bond holdings: +\$1 million	Change in money stock: +\$1 million

- ▶ Assets of the central bank are the bonds it holds in its portfolio.
- ▶ The liabilities are the stock of money in the economy.
- ▶ Open market operation: central bank buys bonds and issues money, increases both assets and liabilities by the same amount.

Bond Prices and Bond Yields

- ▶ Bond prices are determined in bond markets, not interest rates.
- ▶ But there is a direct relationship.
- ▶ Suppose a bond such as a Treasury bill, or T-bill, promises to pay \$100 a year from now (face value).
- ▶ If price of the bond today is $\$P_B$, then interest rate on the bond is:

Financial Intermediaries

- ▶ To understand what determines the interest rate in an economy with both currency and checkable deposits, we must first look at what banks do.
- ▶ Modern economies are characterized by the existence of many types of **financial intermediaries**: institutions that receive funds from people and firms, use these funds to buy financial assets or to make loans.
- ▶ Banks are financial intermediaries that have money, in the form of checkable deposits, as their liabilities.
- ▶ Banks keep as reserves some of the funds they receive: partly in cash, partly in account at central bank.
- ▶ The liabilities of the central bank are the money it has issued, called central bank money.

Three Reasons for Bank Reserves

1. Some depositors withdraw cash from their checking accounts, others deposit
 - ▶ No reason for the inflows and outflows of cash to be equal, so the bank must keep some cash on hand.
2. In the same way, people with accounts at the bank write checks to people with accounts at other banks
 - ▶ What the bank, as a result of these transactions, owes the other banks can be larger or smaller than what the other banks owe to it.
3. In addition, banks are typically subject to reserve requirements,
 - ▶ Require them to hold reserves in some proportion of their checkable deposits.

Balance Sheets

- ▶ Loans represent $\sim 70\%$ of banks' non reserve assets, bonds 30%.
- ▶ The liabilities of the central bank are the money it has issued, called central bank money.

(a)

Central Bank	
Assets	Liabilities
Bonds	Central Bank Money = Reserves + Currency

(b)

Banks	
Assets	Liabilities
Reserves Loans Bonds	Checkable deposits

Demand for Central Bank Money

- ▶ Assume people hold no currency so the demand for money is the demand for checkable deposits, as before:

$$\ominus$$
$$M^d = \$YL(i)$$

- ▶ Hold more checkable deposits with higher level of transactions and lower interest rate on bonds.
- ▶ The larger the amount of checkable deposits, the larger the amount of reserves banks must hold (precautionary, regulatory).
- ▶ Demand for reserves by banks depends on the amount of checkable deposits:

$$H^d = \theta M^d = \theta \$YL(i)$$

- ▶ θ : reserve ratio; H^d : demand for high-powered money (monetary base).

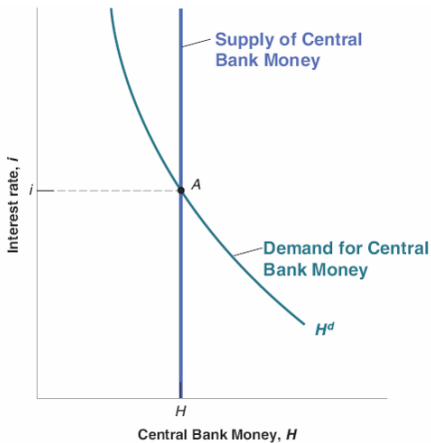
Equilibrium in the Market for Central Bank Money

- ▶ Let H denote the supply of central bank money, then the equilibrium condition:

$$H = H^d \Rightarrow H = \theta Y L(i)$$

- ▶ An increase in H leads to a decrease in the interest rate.
- ▶ The equilibrium interest rate is such that the supply of central bank money is equal to the demand for central bank money.
- ▶ The federal funds market is an actual market for bank reserves.
- ▶ The federal funds rate is the interest rate determined in the federal funds market; main indicator of US monetary policy because the Fed can choose the federal funds rate it wants by changing H .

Determining the Interest Rate



Reserve Requirement in China

- e.g., <http://www.scmp.com/news/china/economy/article/2142146/chinas-central-bank-cuts-reserve-requirement-lenders-free->



SOURCE: TRADINGECONOMICS.COM | THE PEOPLES BANK OF CHINA

Reserve Requirement in China (con.)

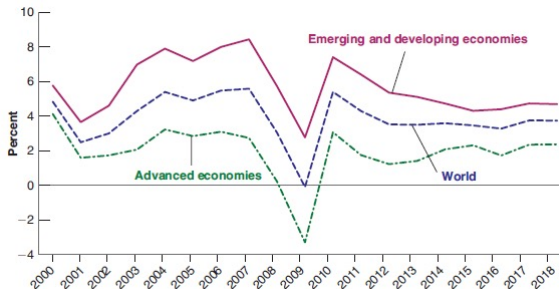


SOURCE: TRADINGECONOMICS.COM | THE PEOPLES BANK OF CHINA

Global Financial Crisis

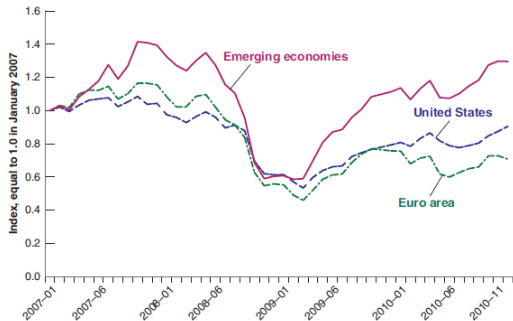
- ▶ From 2000 to 2007, the world economy had a sustained expansion.
- ▶ In 2007, U.S. housing prices started declining, leading to a major financial crisis, then major economic crisis with falling stock prices.
- ▶ The financial crisis turned into a major economic crisis with falling stock prices.
- ▶ In the third quarter of 2008, U.S. output growth turned negative and remained so in 2009.
- ▶ Through the trade and financial channels, the U.S. crisis quickly became a world crisis.

Output growth rates



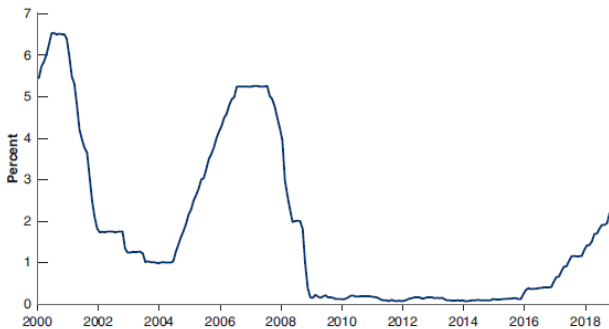
Source: IMF. World Economic Outlook Database, July 2018. NGDP_RPCH.A.

Stock prices 2007-2010



Source: IMF. World Economic Outlook Database, July 2018. NGDP_RPCH.A.

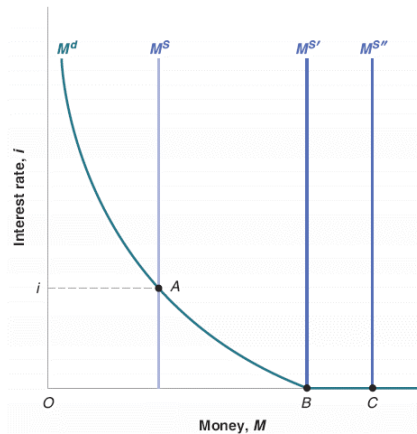
Federal Funds Rate



The Liquidity Trap

- ▶ **Zero lower bound:** The interest rate cannot go below zero.
- ▶ The economy is in a **liquidity trap** when the interest rate is down to zero, monetary policy cannot decrease it further.
- ▶ When the interest rate is equal to zero, and once people have enough money for transaction purposes, they become indifferent between holding money and holding bonds.
- ▶ The demand for money becomes horizontal.
- ▶ This implies that, when the interest rate is equal to zero, further increases in the money supply have no effect on the interest rate, which remains equal to zero.

The Liquidity Trap (con.)



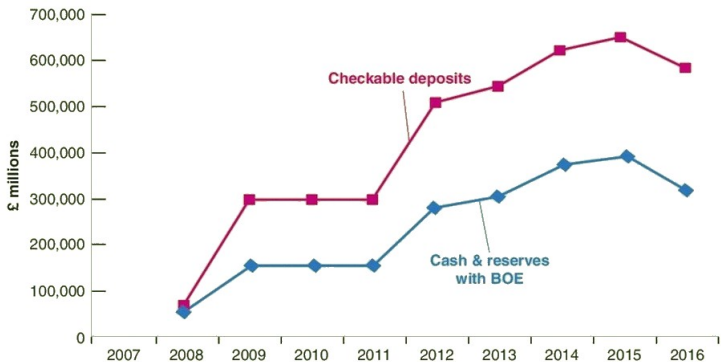
Liquidity Trap in the United Kingdom

- ▶ Global Financial Crisis prompted central banks in many countries to cut short-term policy rates to near zero levels.
- ▶ Monetary Policy Committee of the Bank of England (BoE) cut the bank rate from 5% in mid-2008 to 0.5% in early 2009.
- ▶ In 2015-16, the slowdown in the global economy, coupled with persistently low rates of inflation, has led several central banks across Europe, as well as in Japan, to cut interest rates below zero!
- ▶ BoE was reluctant to cut interest rates below zero in the fear that negative rates could discourage depositors from keeping their savings in banks.
- ▶ The weaknesses of the recovery from the Global Financial Crisis and the European sovereign crisis have led to a decline in household spending and business investment.

Liquidity Trap in the United Kingdom (con.)

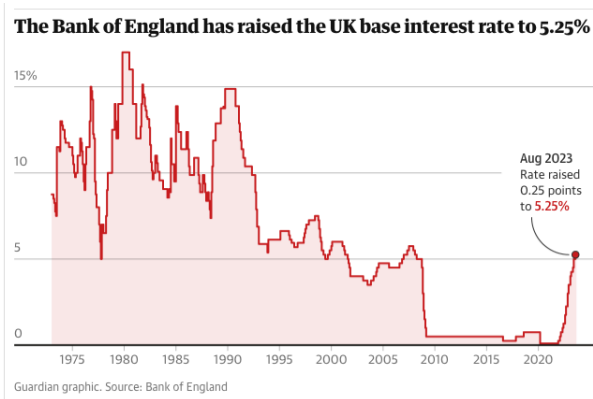
- ▶ Uncertainty in the economic outlook should cause households to increase bank deposits and would prompt banks to increase their reserve holdings, which is exactly what happened.
- ▶ BoE has expanded cash and bank reserves substantially. Bank deposits increased more than ten-fold, from £23.6 billion in 2007 to £265.5 billion in mid-2016.
- ▶ Sharp expansion in money was absorbed by households and by banks as the bank rate remained at 0.5%.

Checkable Deposits and Reserves with BoE, 2007-16



<https://www.nytimes.com/2021/02/04/business/bank-of-england-negative-interest-rates.html>

Interest Rates in UK



[https://www.theguardian.com/business/2023/aug/03/
bank-of-england-raises-uk-interest-rate-to-fresh-15-year-h](https://www.theguardian.com/business/2023/aug/03/bank-of-england-raises-uk-interest-rate-to-fresh-15-year-h)